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(54) **SHORT AND SMALL EXTERNAL DUAL-SPRING SPRAYER**

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Related U.S. Application Data

(63) Continuation of application No. PCT/CN2023/113043, filed on Aug. 15, 2023.

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B05B 11/10 (2023.01)

(52) **U.S. Cl.**
CPC ***B05B 11/1074*** (2023.01); ***B05B 11/1069*** (2023.01); *B05B 11/1073* (2023.01)

(58) **Field of Classification Search**
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USPC 222/321.7, 321.8, 321.9, 320, 321.1, 222/321.2, 321.6; 239/329, 331, 333
See application file for complete search history.

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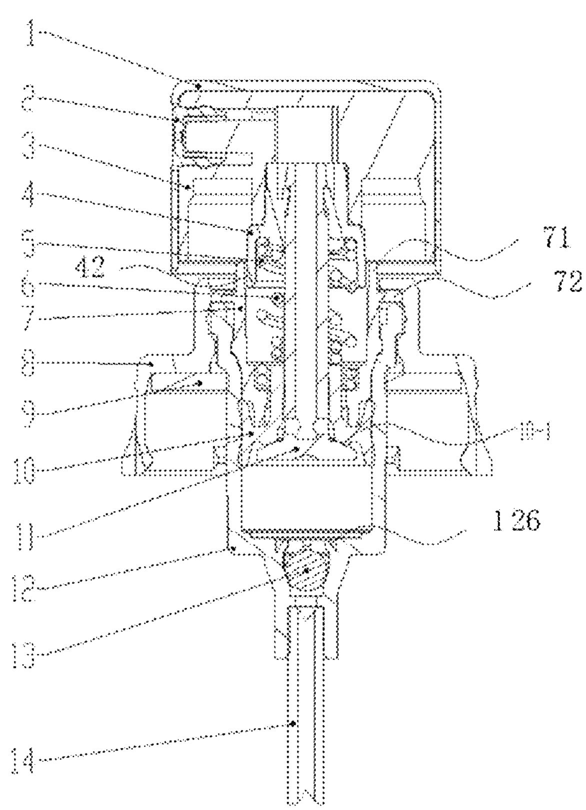
(57) **ABSTRACT**

A short and small external dual-spring sprayer, including a head cap cover, where a head cap is arranged in the head cap cover, a spray piece is arranged at a front end of the head cap, the head cap is connected to a valve rod, the valve rod is sleeved at an upper part of a valve needle, the valve needle penetrates through a bushing and a piston and is further sleeved with an internal spring, the internal spring has an upper part located in the valve rod and a lower end in contact with the piston, an external spring is arranged on an outer side of the internal spring and has an upper part located in a middle sleeve and a lower part located in the bushing.

10 Claims, 9 Drawing Sheets



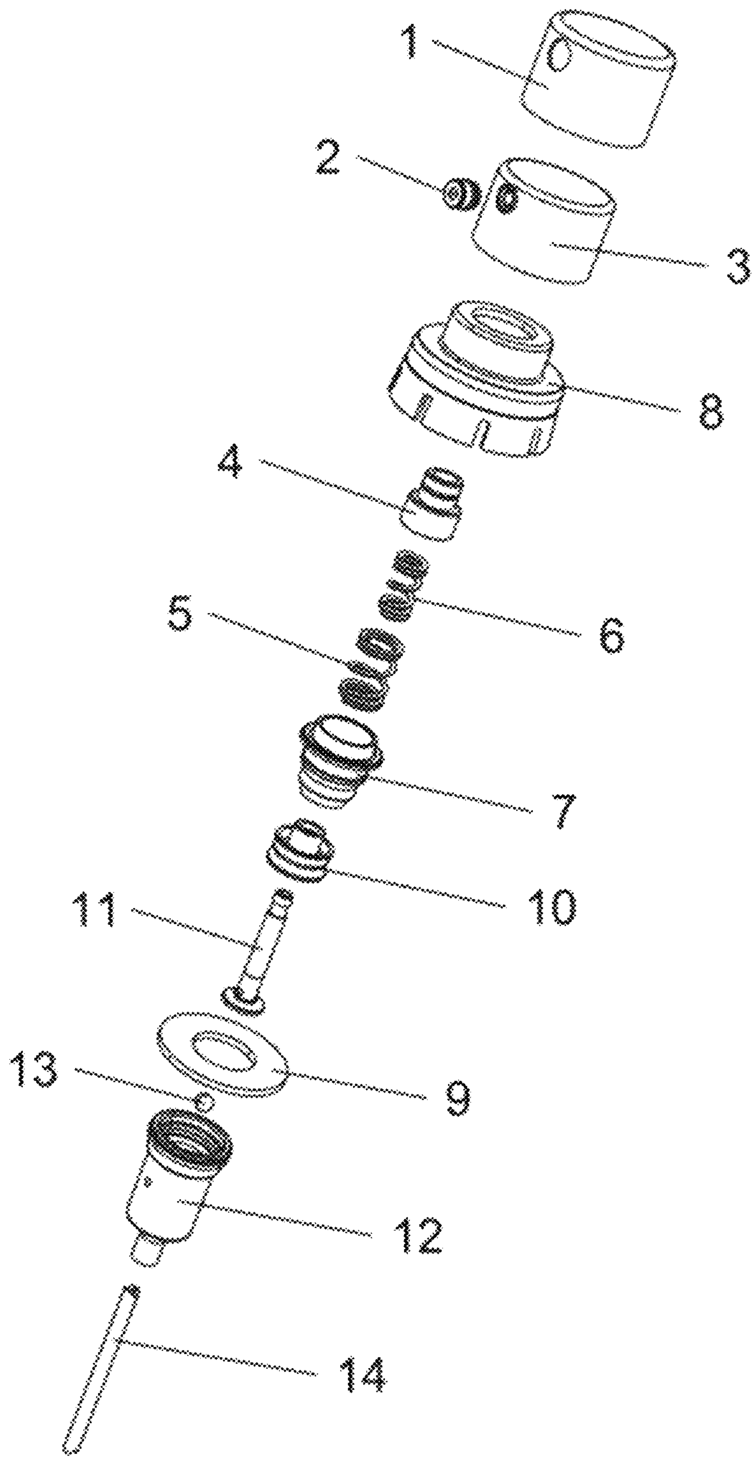


FIG. 1

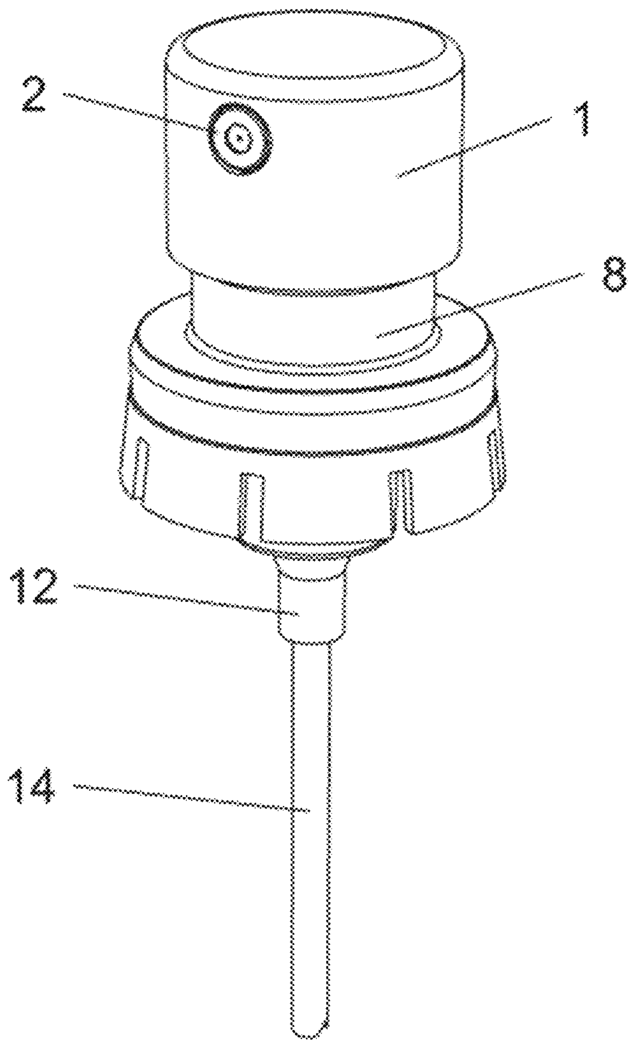


FIG. 2

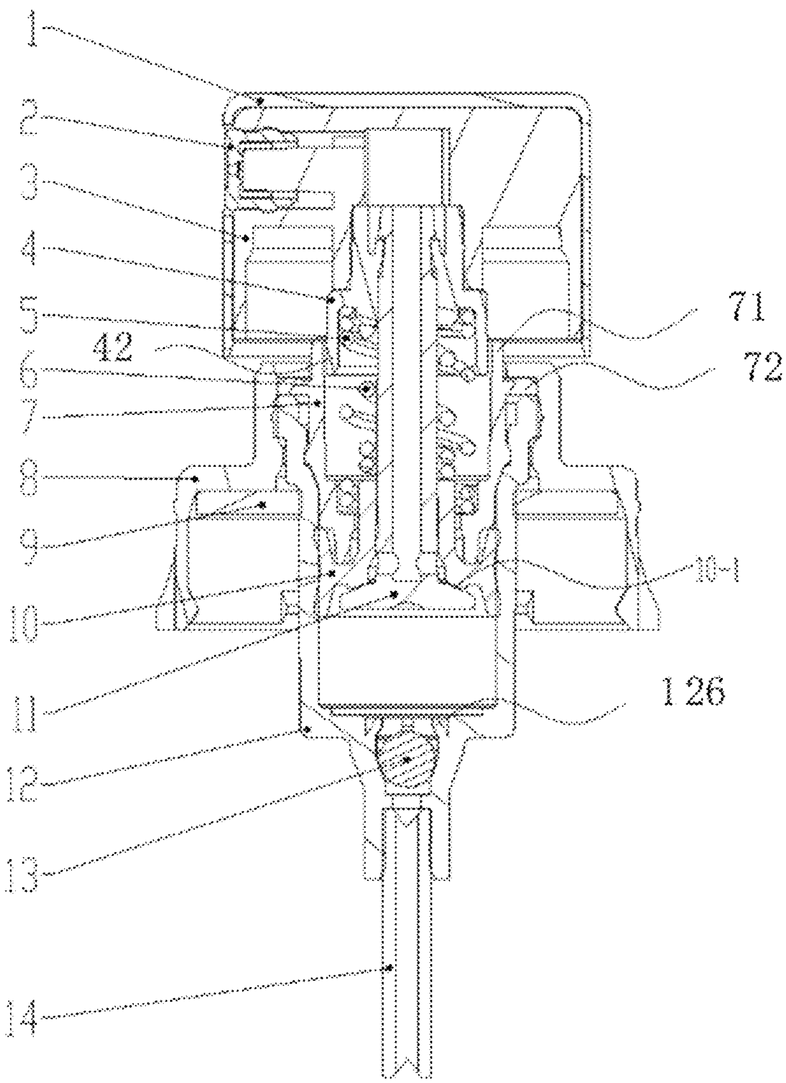


FIG. 3

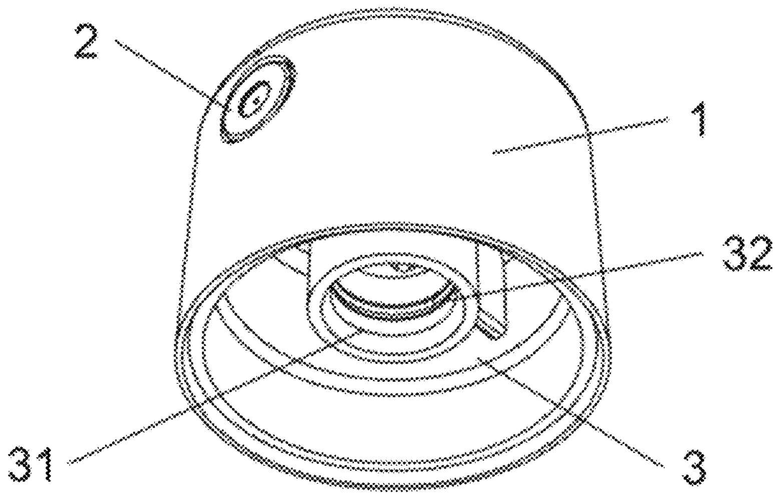


FIG. 4

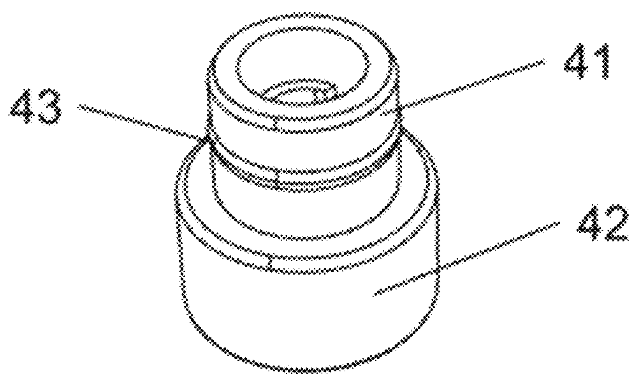


FIG. 5

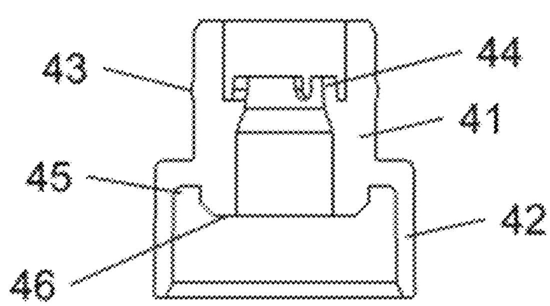


FIG. 6

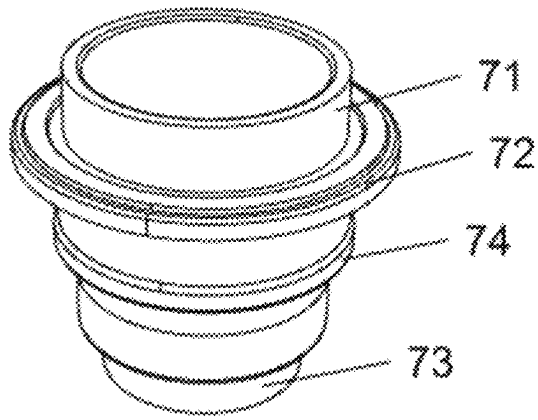


FIG. 7

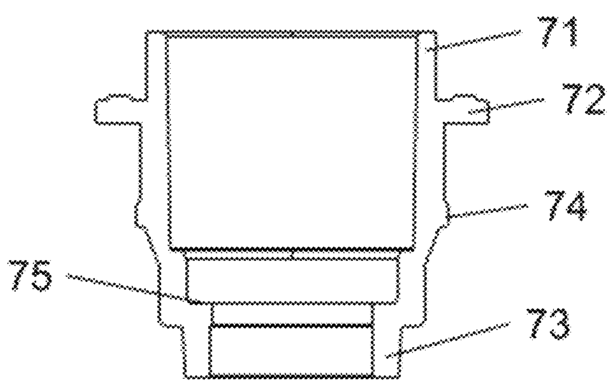


FIG. 8

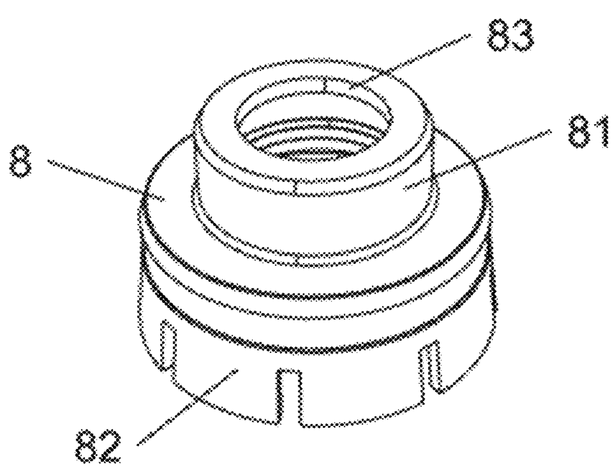


FIG. 9

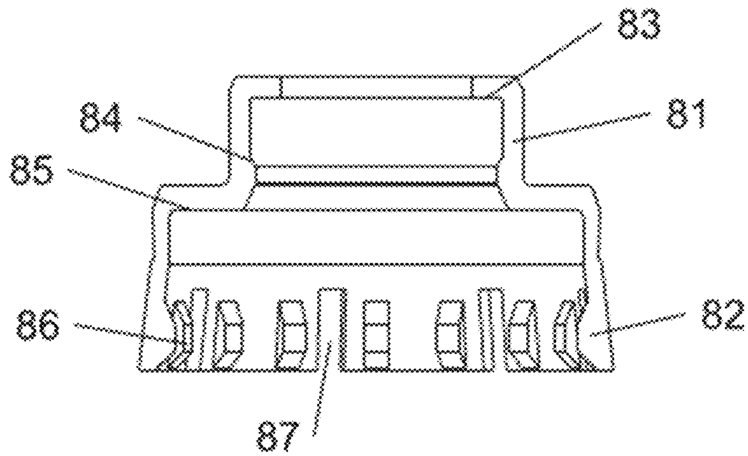


FIG. 10

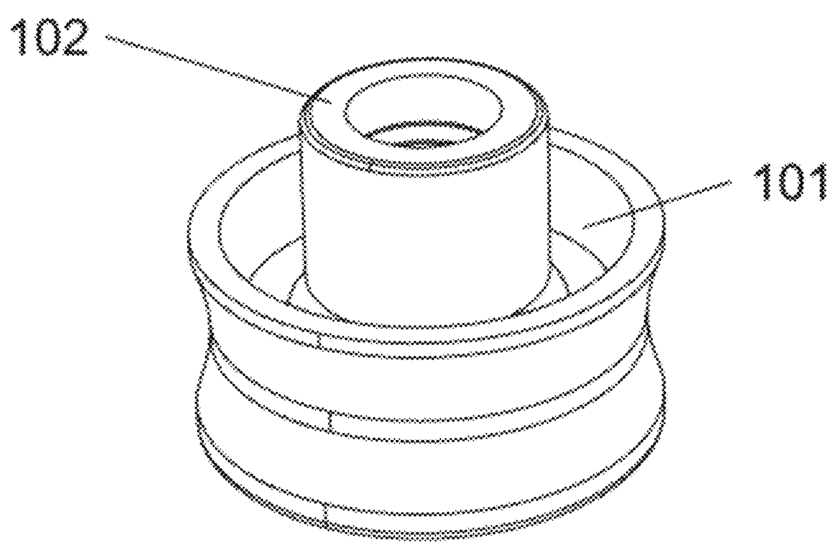


FIG. 11

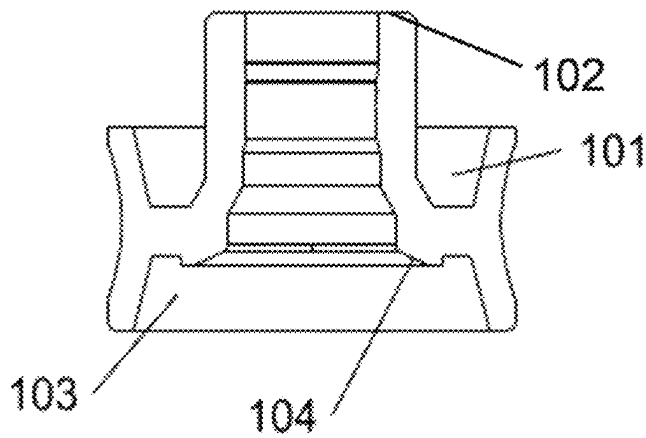


FIG. 12

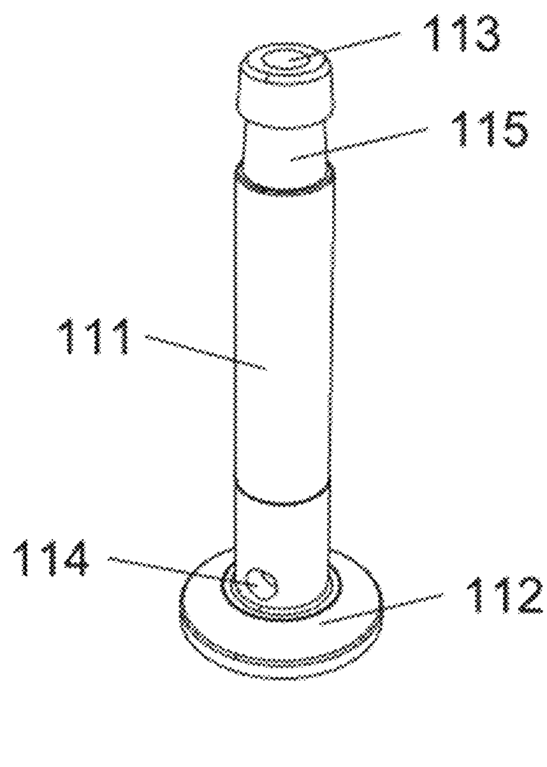


FIG. 13

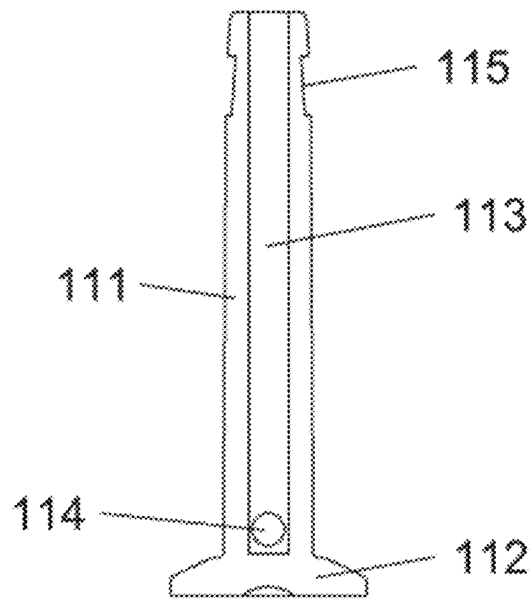


FIG. 14

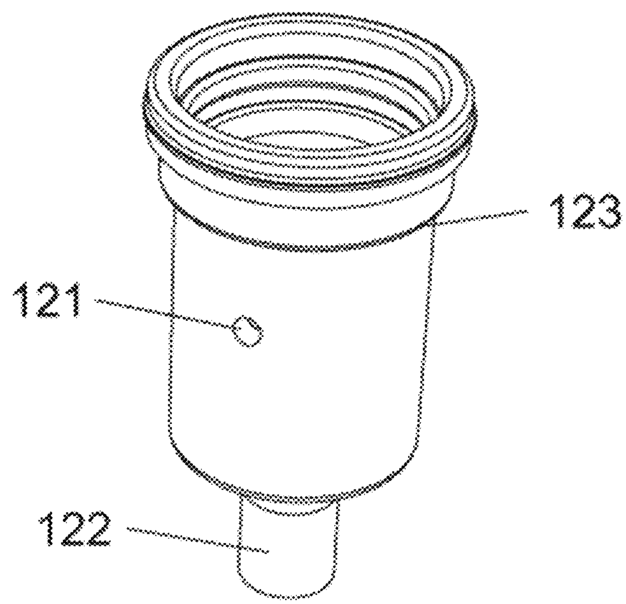


FIG. 15

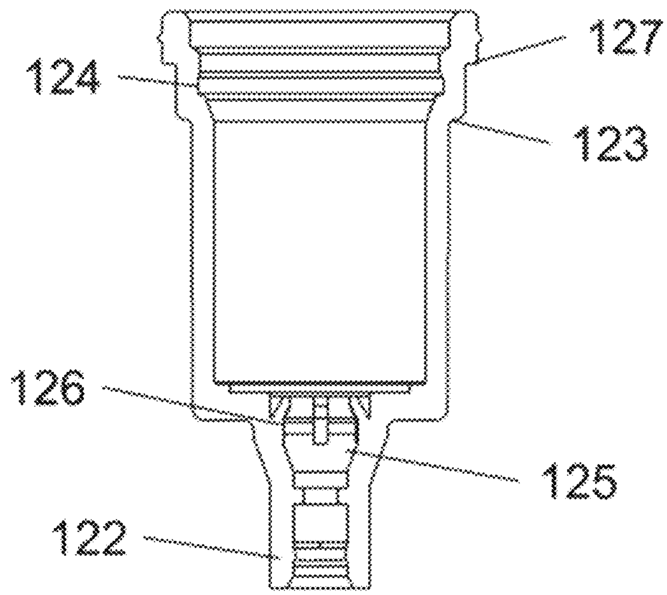


FIG. 16

SHORT AND SMALL EXTERNAL DUAL-SPRING SPRAYER

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of International Application No. PCT/CN2023/113043, with an international filing date of Aug. 15, 2023, the entire contents of all of which are incorporated herein by reference.

FIELD OF TECHNOLOGY

The present invention relates to the technical field of spray pumps, in particular to a short and small external dual-spring sprayer.

BACKGROUND

The pump, a machine configured to increase the pressure of liquid or gas and make it conveyed and flow, is an apparatus configured to move liquid, gas, or a special fluid medium, that is, a machine that does work on a fluid. Pumps are mainly configured to transport liquids including water, oil, acid and alkali liquids, emulsions, suspended emulsions, and liquid metals, and can also transport liquids, gas mixtures, and liquids containing suspended solids.

At present, metal springs are used in most of spray pumps for daily chemical products and/or cleaning products and cosmetic products sold in the market. However, the metal springs are prone to being in contact with contents to cause adverse chemical reactions, resulting in pollution of the contents.

SUMMARY

To solve the above technical problems, the present invention provides a short and small external dual-spring sprayer, including a head cap cover, where a head cap is arranged in the head cap cover, a spray piece is arranged at a front end of the head cap, the head cap is connected to a valve rod, the valve rod is sleeved at an upper part of a valve needle, the valve needle penetrates through a bushing and a piston and is further sleeved with an internal spring, the internal spring has an upper part located in the valve rod and a lower end in contact with the piston, an external spring is arranged on an outer side of the internal spring and has an upper part located in a middle sleeve and a lower part located in the bushing, the middle sleeve is sleeved on an outer side of the bushing and connected to an air cylinder, a gasket is arranged inside the middle sleeve and sleeved on an outer side of the air cylinder, a bottom of the air cylinder is connected to a suction tube, and a steel ball is arranged inside the air cylinder.

Preferably, a sleeving hole is formed inside the head cap, and an annular clamping groove is formed in an inner wall of the sleeving hole.

Preferably, the valve rod includes a fixing portion at an upper part and a support portion at a lower part, an outer diameter of the fixing portion is less than an outer diameter of the support portion, a convex rib is arranged on an outer wall of the fixing portion, a clamping sleeve is arranged inside the fixing portion, an annular external spring upper accommodating groove is formed inside the support portion, and an internal spring upper support region is arranged at a position, located on an inner side of the external spring upper accommodating groove, of the valve rod.

Preferably, a bulge portion is arranged at an upper part of the bushing, a first convex ring and a second convex ring are arranged in a middle of the bushing, an insertion portion is arranged at a bottom of the bushing, and an external spring lower accommodating groove is formed in an inner wall of the bushing.

Preferably, the middle sleeve includes an upper sleeve and a lower sleeve, a first sealing groove is formed in a top wall of the upper sleeve, a support bulge is further arranged in the upper sleeve, and a second sealing groove is formed in a top wall of the lower sleeve and configured to mount the gasket.

Preferably, a middle sleeve bulge arranged annularly is arranged on an inner wall of a bottom of the lower sleeve, and a plurality of grooves are further formed in the bottom of the lower sleeve.

Preferably, an internal spring lower support region is arranged at an upper end of the piston, an annular upper groove is formed in the upper end of the piston, a lower groove is formed in a lower end of the piston, a bevel edge is arranged on an inner wall of the lower groove, the piston is mounted inside the air cylinder, and the piston and the air cylinder form a cavity structure for accommodating a liquid.

Preferably, the valve needle includes a needle body and a sealing ring, an upper end surface of the sealing ring is an inclined surface, a liquid outlet is formed inside the needle body, a liquid inlet is formed in a bottom of the needle body, and an annular groove is formed in an outer wall of an upper part of the needle body.

Preferably, an air pressure balance hole is formed in an outer wall of the air cylinder, a tube connector is arranged at the bottom of the air cylinder, an accommodating cavity is formed in a position, located above the tube connector, of the air cylinder and configured to place the steel ball, and a plurality of inclined ribs are annularly arranged at an upper part of the accommodating cavity.

Preferably, a second external convex rib is arranged on an outer wall of an upper part of the air cylinder, a first external convex rib is arranged at a position, located below the second external convex rib, of the outer wall of the upper part of the air cylinder, and an inner groove is further formed inside the upper part of the air cylinder and configured to be clamped with the second convex ring.

The present invention has the following technical effects and advantages:

1. In the present invention, the internal spring upper support region is arranged at the position, located on the inner side of the external spring upper accommodating groove, of the valve rod, and when mounted, the valve rod is inserted into the bushing, thereby avoiding the inclination of the valve rod.

2. In the present invention, the first convex ring is arranged in the middle of the bushing, an outer diameter of the first convex ring is greater than an outer diameter of the bugle portion, and the first convex ring is attached to a top surface of the middle sleeve, to improve the sealing property.

3. In the present invention, the needle body is connected to the valve rod, and an inclination angle of the inclined surface of the sealing ring is the same as that of the bevel edge, such that the sealing ring can be tightly attached to the bevel edge, thereby achieving the sealing effect.

4. In the present invention, the plurality of inclined ribs are annularly arranged at the upper part of the accommodating cavity, thereby preventing the steel ball from falling off from the accommodating cavity.

5. In the sprayer, a short and small dual-spring structure is designed for attractive appearance of a product, and a metal spring is placed outside the valve needle, thereby

avoiding adverse chemical reactions caused by contact between contents and the metal spring.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is an exploded view of a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **2** is a schematic structural diagram of a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **3** is a sectional view of a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **4** is a schematic structural diagram of a head cap in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **5** is a schematic structural diagram of a valve rod in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **6** is a sectional view of a valve rod in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **7** is a schematic structural diagram of a bushing in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **8** is a sectional view of a bushing in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **9** is a schematic structural diagram of a middle sleeve in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **10** is a sectional view of a middle sleeve in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **11** is a schematic structural diagram of a piston in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **12** is a sectional view of a piston in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **13** is a schematic structural diagram of a valve needle in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **14** is a sectional view of a valve needle in a short and small external dual-spring sprayer provided by an embodiment of the present application;

FIG. **15** is a schematic structural diagram of an air cylinder in a short and small external dual-spring sprayer provided by an embodiment of the present application; and

FIG. **16** is a sectional view of an air cylinder in a short and small external dual-spring sprayer provided by an embodiment of the present application.

In the drawings: **1**. head cap cover; **2**. spray piece; **3**. head cap; **31**. sleeving hole; **32**. clamping groove; **4**. valve rod; **41**. fixing portion; **42**. support portion; **43**. convex rib; **44**. clamping sleeve; **45**. external spring upper accommodating groove; **46**. internal spring upper support region; **5**. external spring; **6**. internal spring; **7**. bushing; **71**. bulge portion; **72**. first convex ring; **73**. insertion portion; **74**. second convex ring; **75**. external spring lower accommodating groove; **8**. middle sleeve; **81**. upper sleeve; **82**. lower sleeve; **83**. first sealing groove; **84**. support bulge; **85**. second sealing groove; **86**. middle sleeve bulge; **87**. groove; **9**. gasket; **10**. piston; **101**. upper groove; **102**. internal spring lower support region; **103**. lower groove; **104**. bevel edge; **11**. valve needle; **111**. needle body; **112**. sealing ring; **113**. liquid outlet; **114**. liquid inlet; **115**. annular groove; **12**. air cylinder; **121**. air pressure balance hole; **122**. tube connector; **123**. first external convex rib; **124**. inner groove; **125**. accommodating cavity; **126**. inclined rib; **127**. second external convex rib; **13**. steel ball; and **14**. suction tube.

DESCRIPTION OF THE EMBODIMENTS

The present invention is further described in detail below in conjunction with the accompanying drawings and the specific embodiments. The embodiments of the present invention are provided for examples and description, and are not exhaustive or limit the present invention to the disclosed form. Many modifications and variations are obvious to those of ordinary skill in the art. The embodiments are selected and described to better illustrate the principles and practical applications of the present invention, and to enable those of ordinary skill in the art to understand the present invention and thus design various embodiments with various modifications suitable for specific purposes.

As shown in FIGS. **1-3**, a short and small external dual-spring sprayer is provided in this embodiment, including a head cap cover **1**, where a head cap **3** is arranged in the head cap cover **1**, a spray piece **2** is arranged at a front end of the head cap **3** and configured to spray a liquid, the head cap cover **1** is provided with a through hole corresponding to the spray piece **2**, the head cap **3** is connected to a valve rod **4**, the valve rod **4** is sleeved at an upper part of a valve needle **11**, the valve needle **11** penetrates through a bushing **7** and a piston **10** and is further sleeved with an internal spring **6**, the internal spring **6** has an upper part located in the valve rod **4** and a lower end in contact with the piston **10**, an external spring **5** is arranged on an outer side of the internal spring **6** and has an upper part located in a middle sleeve **8** and a lower part located in the bushing **7**, the middle sleeve **8** is sleeved on an outer side of the bushing **7** and connected to an air cylinder **12**, a gasket **9** is arranged inside the middle sleeve **8** and sleeved on an outer side of the air cylinder **12**, a sealing effect is achieved when the middle sleeve **8** is connected to a bottle body through the arrangement of the gasket **9**, a bottom of the air cylinder **12** is connected to a suction tube **14**, and a steel ball **13** is arranged inside the air cylinder **12**. In the sprayer, a dual-spring structure is arranged, and a metal spring is placed outside the valve needle **11**, thereby avoiding adverse chemical reactions caused by contact between contents and the metal spring.

Further, in this embodiment, as shown in FIG. **4**, a sleeving hole **31** is formed inside the head cap **3**, an annular clamping groove **32** is formed in an inner wall of the sleeving hole **31**, the sleeving hole **31** communicates with the spray piece **2** and is configured to be inserted with the valve rod **4**, and the clamping groove **32** is clamped with the valve rod **4** in a matched manner to connect the valve rod **4** to the sleeving hole **31**.

Further, in this embodiment, as shown in FIGS. **5-6**, the valve rod **4** is hollow and includes a fixing portion **41** at an upper part and a support portion **42** at a lower part, an outer diameter of the fixing portion **41** is less than an outer diameter of the support portion **42**, a convex rib **43** is arranged on an outer wall of the fixing portion **41** and is embedded into the clamping groove **32** to fix the valve rod **4** and the sleeving hole **31** when the valve rod **4** is inserted into the sleeving hole **31**, a clamping sleeve **44** is arranged inside the fixing portion **41**, the valve needle **11** penetrates through the clamping sleeve **44** from the bottom up along the valve rod **4**, the valve needle **11** and the valve rod **4** are fixed by the clamping sleeve **44**, an annular external spring upper accommodating groove **45** is formed inside the support portion **42**, an upper part of the external spring **5** is embedded into the external spring upper accommodating groove **45**, an internal spring upper support region **46** is arranged at a position, located on an inner side of the external spring upper accommodating groove **45**, of the valve rod **4**, and when mounted, the valve rod **4** is inserted into the bushing **7**, thereby avoiding the inclination of the valve rod **4**.

Further, in this embodiment, as shown in FIGS. **7-8**, the bushing **7** is hollow, a bulge portion **71** is arranged at an upper part of the bushing **7**, an outer wall of the bulge portion **71** is a smooth surface, the bulge portion **71** penetrates through the middle sleeve **8** from the bottom up, and an upper end surface of the bulge portion **71** is higher than the middle sleeve **8**, such that when the head cap **3** is pressed, the valve rod **4** is kept vertical; a first convex ring **72** is arranged in a middle of the bushing **7**, an outer diameter of the first convex ring **72** is greater than an outer diameter of the bulge portion **71**, and the first convex ring **72** is attached to a top surface of the middle sleeve **8**, to improve the sealing property; an insertion portion **73** is arranged at a bottom of the bushing **7**, an outer diameter of the insertion portion **73** is less than the outer diameter of the bulge portion **71**, and the insertion portion **73** is configured to be inserted into the piston **10**; a second convex ring **74** is further arranged in the middle of the bushing **7**, and an outer diameter of the second convex ring **74** is less than the outer diameter of the first convex ring **72**; the bushing **7** is inserted into the air cylinder **12**, and the second convex ring **74** is in contact with an interior of the air cylinder **12** to achieve the sealing effect; and an external spring lower accommodating groove **75** is formed in an inner wall of the bushing **7**, and a bottom of the external spring **5** is located in the external spring lower accommodating groove **75** to support the external spring **5**.

In this embodiment, as shown in FIGS. **9-10**, the middle sleeve **8** is hollow and includes an upper sleeve **81** and a lower sleeve **82**, an outer diameter of the lower sleeve **82** is greater than an outer diameter of the upper sleeve **81**, the upper sleeve **81** is configured to be inserted into the bulge portion **71**, a first sealing groove **83** is formed in a top wall of the upper sleeve **81** and configured to be tightly attached to an upper surface of the first convex ring **72** to achieve the sealing effect, an annular support bulge **84** is further arranged in the upper sleeve **81**, a second sealing groove **85** is formed in a top wall of the lower sleeve **82** and configured to mount the gasket **9**, a middle sleeve bulge **86** arranged annularly is arranged on an inner wall of a bottom of the lower sleeve **82** and configured to be connected to a bottle mouth, a plurality of grooves **87** are further formed in the bottom of the lower sleeve **82**, the lower sleeve **82** is sleeved on the bottle mouth, and the bottom of the lower sleeve **82** is prone to deformation through the arrangement of the grooves **87**, which facilitates the lower sleeve **82** to be sleeved on the bottle mouth.

In this embodiment, as shown in FIGS. **11-12**, the piston **10** is hollow, an internal spring lower support region **102** is arranged at an upper end of the piston **10**, an annular upper groove **101** is formed in the upper end of the piston **10** and configured to be embedded into the insertion portion **73**, a lower groove **103** is formed in a lower end of the piston **10**, an annular bevel edge **104** is arranged on an inner wall of the lower groove **103** and configured to match with the valve needle **11** to achieve the sealing effect, the piston **10** is mounted inside the air cylinder **12** and can slide along the air cylinder **12**, and the piston **10** and the air cylinder **12** form a cavity structure for accommodating a liquid.

In this embodiment, as shown in FIGS. **13-14**, the valve needle **11** includes a needle body **111** and a sealing ring **112**, an upper end surface of the sealing ring **112** is an inclined surface, a liquid outlet **113** is formed inside the needle body **111**, and a liquid inlet **114** is formed in a bottom of the needle body **111**, such that the liquid can enter the liquid inlet **114** and flow out of the liquid outlet **113**; an annular groove **115** is formed in an outer wall of an upper part of the needle body **111**, and when the upper part of the needle body **111** is inserted into the valve rod **4**, the clamping sleeve **44** of the valve rod **4** is embedded into the annular groove **115** to connect the needle body **111** to the valve rod **4**; and an inclination angle of the inclined surface of the sealing ring **112** is the same as that of the bevel edge **104**, such that the sealing ring **112** can be tightly attached to the bevel edge **104**, thereby achieving the sealing effect.

In this embodiment, as shown in FIGS. **15-16**, an air pressure balance hole **121** is formed in an outer wall of the air cylinder **12**, a tube connector **122** is arranged at the bottom of the air cylinder **12** and connected to the suction tube **14**, an accommodating cavity **125** is formed in a position, located above the tube connector **122**, of the air cylinder **12** and configured to place the steel ball **13**, and the steel ball **13** can move up and down along the accommodating cavity **125**; a plurality of inclined ribs **126** are annularly arranged at an upper part of the accommodating cavity **125**, thereby preventing the steel ball **13** from falling off from the accommodating cavity **125**; when the steel ball **13** is located at a bottom of the accommodating cavity **125**, the tube connector **122** is blocked, and the bottom of the air cylinder **12** is in a sealed state; when the steel ball **13** is separated from the bottom of the accommodating cavity **125** under the action of atmospheric pressure, the liquid enters the accommodating cavity **125** from the suction tube **14** and the tube connector **122**, and then flows into the air cylinder **12** from a gap between the steel ball **13** and the accommodating cavity **125**; the liquid is stored in the cavity structure formed by the piston **10** and the air cylinder **12**, waiting for the next liquid discharge; and further, a second external convex rib **127** is arranged on an outer wall of an upper part of the air cylinder **12**, the air cylinder **12** is inserted into the middle sleeve **8** from the bottom up, the second external convex rib **127** is inserted into an upper end of the support bulge **84**, the air cylinder **12** is supported by the support bulge **84** to prevent the air cylinder **12** from falling off from the middle sleeve **8**, a first external convex rib **123** is arranged at a position, located below the second external convex rib **127**, of the outer wall of the upper part of the air cylinder **12** and configured to be attached to an upper surface of the gasket **9** to achieve the sealing effect and prevent the liquid from flowing out of a gap between the middle sleeve **8** and the air cylinder **12**, an inner groove **124** is further formed inside the upper part of the air cylinder **12** and configured to be clamped with the second convex ring **74**, the bushing **7** is inserted into the air cylinder **12**, and the second convex ring **74** is inserted into the inner groove **124** to connect the bushing **7** to the air cylinder **12**, thereby preventing the bushing **7** and the air cylinder **12** from being loosened.

A working principle of the present invention is as follows:

During use, the head cap cover **1** is pressed down, the head cap **3** and the spray piece **2** move down, the head cap **3** squeezes down the valve rod **4** and the valve needle **11**, the valve rod **4** squeezes down the upper ends of the external spring **5** and the internal spring **6**, the lower end of the external spring **5** squeezes the bushing **7** to move down along the middle sleeve **8**, and meanwhile, the lower end of the internal spring **6** squeezes the piston **10** to move down along the air cylinder **12** to squeeze the liquid in the cavity structure; in this case, air enters the air cylinder **12** from the air pressure balance hole **121**, such that air pressure, located in a space above the piston **10**, of the air cylinder **12** is in a balanced state, the steel ball **13** is tightly attached to the bottom of the accommodating cavity **125**, the tube connector **122** is in a sealed state, the liquid will not flow out of the tube connector **122**, a space of the cavity structure formed by the piston **10** and the air cylinder **12** is gradually reduced, the liquid in the cavity structure flows up and squeezes the piston **10** to move up, the piston **10** and the valve needle **11** move relatively, the upper surface of the sealing ring **112** of the valve needle **11** is separated from the bevel edge **104** of the piston **10**, the liquid inlet **114** is exposed, and the liquid flows through the lower groove **103**, enters the liquid outlet **113** from the liquid inlet **114**, then enters the head cap **3**, and is sprayed from the spray piece **2**;

The head cap cover **1** is loosened, the valve rod **4**, the valve needle **11**, and the piston **10** are gradually reset under elastic forces of the external spring **5** and the internal spring **6**, and the space of the cavity structure formed by the piston **10** and the air cylinder **12** is gradually expanded to generate negative pressure; and in this case, the steel ball **13** is jacked up under the action of atmospheric pressure, the liquid in the suction tube **14** enters the cavity structure through the gap between the steel ball **13** and the accommodating cavity **125**, the external spring **5** and the internal spring **6** are reset, and the valve rod **4**, the valve needle **11**, and the piston **10** are reset, and the liquid is stored in the cavity structure, waiting for the next liquid discharge.

Apparently, the described embodiments are merely part rather than all of the embodiments of the present invention. All other embodiments obtained by those of ordinary skill in the art and related fields based on the embodiments of the present invention without creative efforts should fall within the scope of protection of the present invention. The structures, apparatuses and methods of operation not specifically described and explained in the present invention are implemented according to conventional means in the art, unless specifically described and limited.

What is claimed is:

1. A short and small external dual-spring sprayer, comprising a head cap cover, wherein a head cap is arranged in the head cap cover, a spray piece is arranged at a front end of the head cap, the head cap is connected to a valve rod, the valve rod is sleeved at an upper part of a valve needle, the valve needle penetrates through a bushing and a piston and is further sleeved with an internal spring, the internal spring has an upper part located in the valve rod and a lower end in contact with the piston, an external spring is arranged on an outer side of the internal spring and has an upper part located in a middle sleeve and a lower part located in the bushing, the middle sleeve is sleeved on an outer side of the bushing and connected to an air cylinder, a gasket is arranged inside the middle sleeve and sleeved on an outer side of the air cylinder, a bottom of the air cylinder is connected to a suction tube, and a steel ball is arranged inside the air cylinder.

2. The short and small external dual-spring sprayer of claim **1**, wherein a sleeving hole is formed inside the head cap, and an annular clamping groove is formed in an inner wall of the sleeving hole.

3. The short and small external dual-spring sprayer of claim **1**, wherein the valve rod comprises a fixing portion at an upper part and a support portion at a lower part, an outer diameter of the fixing portion is less than an outer diameter of the support portion, a convex rib is arranged on an outer wall of the fixing portion, a clamping sleeve is arranged inside the fixing portion, an annular external spring upper accommodating groove is formed inside the support portion, and an internal spring upper support region is arranged at a position, located on an inner side of the external spring upper accommodating groove, of the valve rod.

4. The short and small external dual-spring sprayer of claim **1**, wherein a bulge portion is arranged at an upper part of the bushing, a first convex ring and a second convex ring are arranged in a middle of the bushing, an insertion portion is arranged at a bottom of the bushing, and an external spring lower accommodating groove is formed in an inner wall of the bushing.

5. The short and small external dual-spring sprayer of claim **4**, wherein a second external convex rib is arranged on an outer wall of an upper part of the air cylinder, a first external convex rib is arranged at a position, located below the second external convex rib, of the outer wall of the upper part of the air cylinder, and an inner groove is further formed inside the upper part of the air cylinder and configured to be clamped with the second convex ring.

6. The short and small external dual-spring sprayer of claim **1**, wherein the middle sleeve comprises an upper sleeve and a lower sleeve, a first sealing groove is formed in a top wall of the upper sleeve, a support bulge is further arranged in the upper sleeve, and a second sealing groove is formed in a top wall of the lower sleeve and configured to mount the gasket.

7. The short and small external dual-spring sprayer of claim **6**, wherein a middle sleeve bulge arranged annularly is arranged on an inner wall of a bottom of the lower sleeve, and a plurality of grooves are further formed in the bottom of the lower sleeve.

8. The short and small external dual-spring sprayer of claim **1**, wherein an internal spring lower support region is arranged at an upper end of the piston, an annular upper groove is formed in the upper end of the piston, a lower groove is formed in a lower end of the piston, a bevel edge is arranged on an inner wall of the lower groove, the piston is mounted inside the air cylinder, and the piston and the air cylinder form a cavity structure for accommodating a liquid.

9. The short and small external dual-spring sprayer of claim **8**, wherein the valve needle comprises a needle body and a sealing ring, an upper end surface of the sealing ring is an inclined surface, a liquid outlet is formed inside the needle body, a liquid inlet is formed in a bottom of the needle body, and an annular groove is formed in an outer wall of an upper part of the needle body.

10. The short and small external dual-spring sprayer of claim **1**, wherein an air pressure balance hole is formed in an outer wall of the air cylinder, a tube connector is arranged at the bottom of the air cylinder, an accommodating cavity is formed in a position, located above the tube connector, of the air cylinder and configured to place the steel ball, and a plurality of inclined ribs are annularly arranged at an upper part of the accommodating cavity.

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